WHAT HAPPENS WHEN YOU DIET?

THE NEED FOR ENERGY

From taking a breath to using your brain, the human body relies on energy to function. This energy primarily comes from carbohydrates (which are broken down into sugar), but also through fats and proteins.

When we eat more calories than needed, the body converts the excess into fat to store for later use. This way, there is always enough stored for times of crisis when food might not be available.

GOING ON A CALORIE DEFICIT

When you start burning more calories or eat less than what's needed, the body does two things to burn fat:

1. The brain signals fat cells to release their fatty acid molecules, which are then broken down and used as energy. The empty fat cells are discarded as urine, carbon dioxide or perspiration.

2. Over time, the body will start using energy from food directly instead of storing it first as fat. This helps improve your baseline metabolic rate (the chemical reactions that create and break down energy).

FASTING

Insulin produced by your pancreas is a key hormone to tell your body when circulating glucose is available in the bloodstream, or when there is none.

When you don't eat for a certain period of time, the level of insulin is less. This signals your brain to start using stored fat. High insulin levels, on the other hand, stall the fat-burning process for stored fats.

unhealthy body image. According to Rajeswari V. Shetty, head of department, Dietetics, SL Raheja Hospital, Mumbai, social media pushes even those who don't need to lose weight to go on calorie-restrictive diets. "There is no goal, no logic, no nutritional aspect to it other than the feelgood factor of being on a diet and possibly losing some weight," says Shetty. A study published in 2021 by Mintel, a market research firm, found 48 per cent of the 3,000 Indians it surveyed to be 'influenced health consumers', or those who take cues from advertisements and social media trends.

"The information is so conflicting," says Khosla. "Getting expert advice is important to separate facts from fads." Can online applications play the role of experts? Even for those who turn to tailored health services provided by online health apps—and their number is rising—things can get challenging. According to MoEngage, an insights-led platform for customer engagement, there was a 46 per cent rise in health and fitness app downloads between the first and second quarters of 2020 globally; in India, the increase was the highest—156 per cent, or 58 million new users. HealthifyMe, a leading Indian health app, saw an increase of 5 million users during the pandemic.

Apart from HealthifyMe, a few other services have also seen fantastic growth. EatFit, for instance, allows people to buy balanced daily meals, and has half a million Indian users. Instead of offering a diet plan, it offers diet-friendly meals such as rice bowls, lentils, vegetables, etc. For busy working Indians, it's an ideal offering. "Our everyday meal/thali segment has 250,000 users and we have an overall retention rate of 30 per cent," says Ankit Nagori, founder and CEO of Curefoods, which houses the brand EatFit. Another app, Keto India, was founded in 2019 by Sahil Pruthi, a former software developer, with a singular focus on Keto, a unique high-fat diet. Later, it expanded its services and renamed itself Livofy, focusing on weight loss, diabetes, thyroid and PCOS patients. "Our nutritionists have an idea about how to build better habits," he says. "So, we don't give people just a diet plan but also the motivation to turn the plan into a habit." Last year, Pruthi was seen on the business reality show Shark Tank India, where he bagged an offer of Rs 1.6 crore for his app.

Users, however, must be judicious while choosing the myriad services being offered online as health becomes a market in itself. A study by the National Institute of Nutrition published in 2019 found that 13 out of the 20 most downloaded weight loss and calorie-counting apps do not align with Indian recommendations. Each of the 20 apps gave different suggestions. Worse, only eight of the 20 apps encouraged healthy practices, such as consuming fruits and vegetables, limiting saturated fatty acids and having high-fibre fruits.

KEEP IT SIMPLE

All this is not to say that one should abjure monitoring their food habits. The science behind weight loss remains quite simple (see *What Happens When You Diet?*). Fat cells store energy. If one consumes more calories than they need, the body stores the excess energy by expanding the number and size of fat cells. As a biological measure for periods when the food intake may be less, the body prioritises storing fat to the extent of depositing fat cells on muscles, the liver and other organs. When you begin to consume fewer calories, the energy required by your body is drawn from your fat reserves. This happens through a network of signals and hormones,